

DataCORE

Descriptive Statistics, Distributions & Visualizing Your Variables

Afternoon: DataCORE

1. Descriptive Statistics & Distributions
2. Visualizing Your Variables — Histogram, Bar, Scatter
3. Scientific Storytelling — Chart Annotations

*Yesterday you computed one number for your DV. Today we ask: what is the **shape** of your data — and how do we **show** it?*

1:00 – 1:50 · Descriptive Statistics & Distributions



One number hides the story

- A **mean** tells you the *center* — but two very different datasets can share the same mean.
- We need numbers for the **center** *and* the **spread**, plus a picture of the **shape**.
- Today's pipeline: **summarise** → **visualize** → **describe in plain English**.



We use `study_habits.csv` — swap in **your** team file.

Big idea: a *distribution* is the full pattern of values a variable takes — not just its average.

Reload the dataset

Python R

```
import pandas as pd
df = pd.read_csv("study_habits.csv") # rows = students, columns = variables
print(df.head())
```

	student_id	grade_level	study_method	hours_of_sleep	test_score
0	1	9th	solo	7.5	82.0
1	2	9th	group	6.0	74.0
2	3	9th	tutor	8.0	88.0
3	4	10th	solo	5.5	68.0
4	5	10th	group	7.0	79.0

 Columns: grade_level, study_method, hours_of_sleep, test_score.

Mean, median, standard deviation

Python R

```
dv = df["test_score"]  
print("mean :", round(dv.mean(), 2))      # the average
```

mean : 80.42

```
print("median:", dv.median())              # the middle value
```

median: 80.0

```
print("std :", round(dv.std(), 2))        # typical distance from the mean
```

std : 9.47

 `df.describe()` (pandas) / `summary(df)` (R) gives all of these at once.

☑️☰ Categorical stats: frequencies and percentages

Python R

```
cat_counts = df["study_method"].value_counts(dropna=False).rename("frequency")
cat_percent = (df["study_method"].value_counts(dropna=False, normalize=True) * 100).round(1).rename("percent")
cat_table = pd.concat([cat_counts, cat_percent], axis=1)
print(cat_table)
```

	frequency	percent
study_method		
solo	7	35.0
group	7	35.0
tutor	6	30.0

i Use this for any categorical variable (e.g., `grade_level`, `study_method`) to report how many observations are in each group and what share of the sample each group represents.

What each number tells you

Mean

The balance point. Uses *every* value
— so it gets **pulled by outliers**.

Median

The middle value. **Robust to outliers** — better when data is skewed.

Std. Dev.

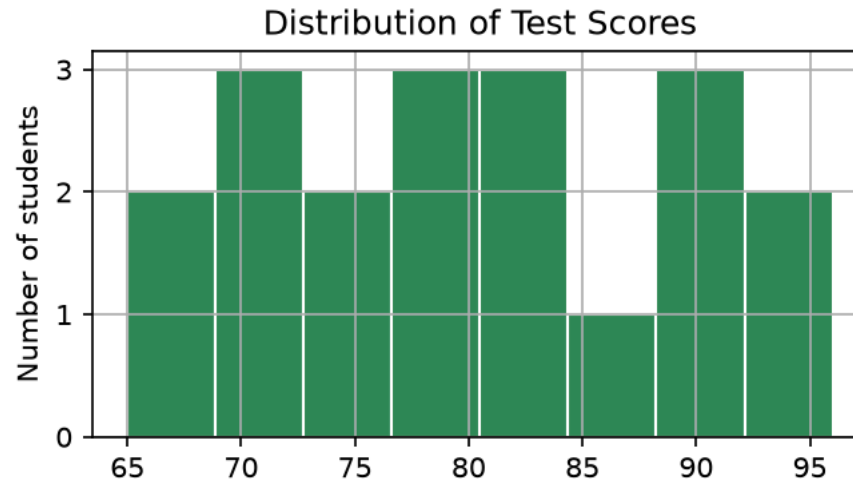
Typical distance from the mean.
Small = clustered; **large** = spread out.

 **When is median better than mean?** Income, house prices, test scores with a few extreme values
— one billionaire (or one zero) drags the mean, not the median.

df.hist() – the shape of your distribution

Python R

```
df["test_score"].hist(bins=8, color="#2e8b57", edgecolor="white")  
plt.title("Distribution of Test Scores")  
plt.xlabel("Test score"); plt.ylabel("Number of students")  
plt.show()
```



 **Read the shape:** is it *normal* (bell), *skewed* (a long tail), or *bimodal* (two humps)?

Three shapes you will meet

Normal

Symmetric bell. Mean $\approx$ median.
Most values near the center.

Skewed

One long tail. Mean is pulled toward the tail — **report the median**.

Bimodal

Two peaks. Often a sign of **two hidden groups** mixed together.

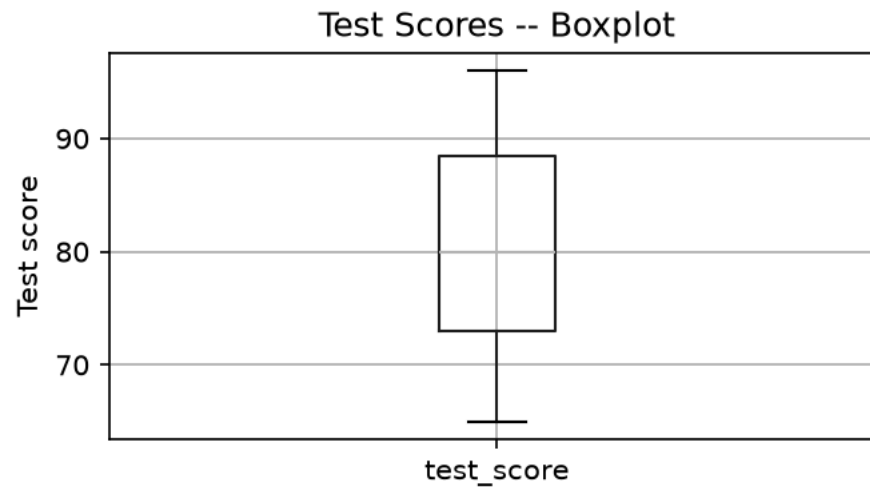


A bimodal DV is a clue: maybe boys vs. girls, or tutored vs. solo studiers behave differently.

df.boxplot() – spot outliers visually

Python R

```
df.boxplot(column="test_score")  
plt.title("Test Scores -- Boxplot")  
plt.ylabel("Test score")  
plt.show()
```



- ⦿ The **box** holds the middle 50%; the **line** is the median; **dots** beyond the whiskers are possible outliers.

Team Pipeline

Run the pipeline on your DV – ~15 min

On your team's dependent variable, record:

min	max	mean	SD
?	?	?	?

Then run `df.hist()` **and** `df.boxplot()` on it.

 Record the four numbers **and** your sentence on your Mission Sheet.

Write ONE sentence describing the distribution in plain English:

*"Test scores ranged from 65 to 96, centered around 81, and were roughly **bell-shaped** with no extreme outliers."*

1:50 – 2:45 · Visualizing Your Variables

Match the chart to the variable



Histogram

One *continuous* variable.
Shows its distribution.



Bar chart

One *categorical* variable.
Compares groups.



Scatter plot

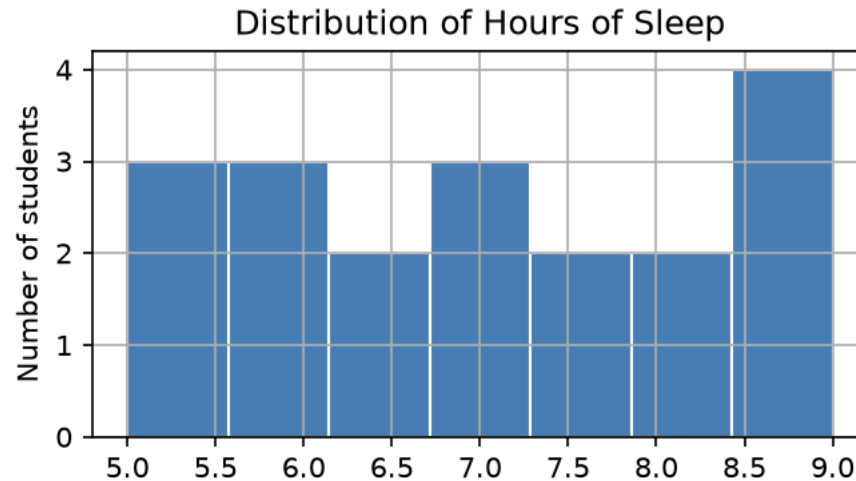
Two continuous variables.
Shows their relationship.

 Instructor demos one; you replicate all three for **your** variables.

Histogram – a continuous variable

Python R

```
df["hours_of_sleep"].hist(bins=7, color="#4a7fb5", edgecolor="white")  
plt.title("Distribution of Hours of Sleep")  
plt.xlabel("Hours of sleep"); plt.ylabel("Number of students")  
plt.show()
```

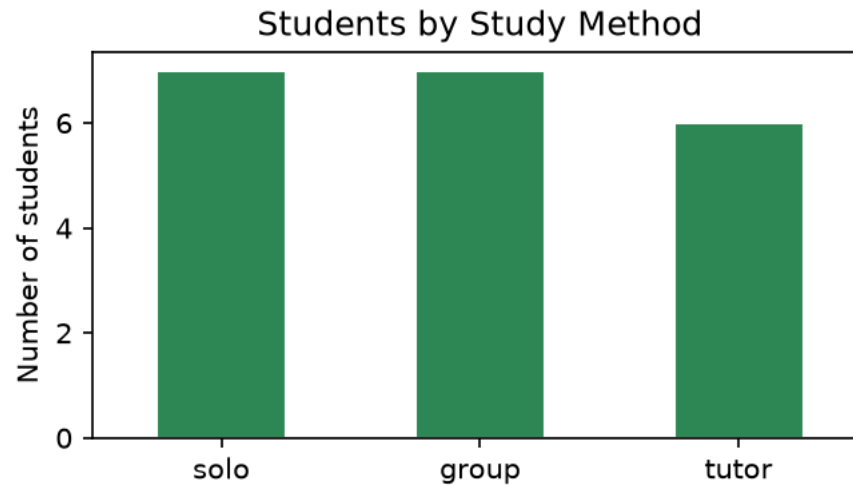


 **Bins** group the values — too few hides shape, too many makes it noisy.

Bar chart – a categorical variable

Python R

```
counts = df["study_method"].value_counts()
counts.plot(kind="bar", color="#2e8b57", edgecolor="white")
plt.title("Students by Study Method")
plt.xlabel("Study method"); plt.ylabel("Number of students")
_ = plt.xticks(rotation=0)
plt.show()
```

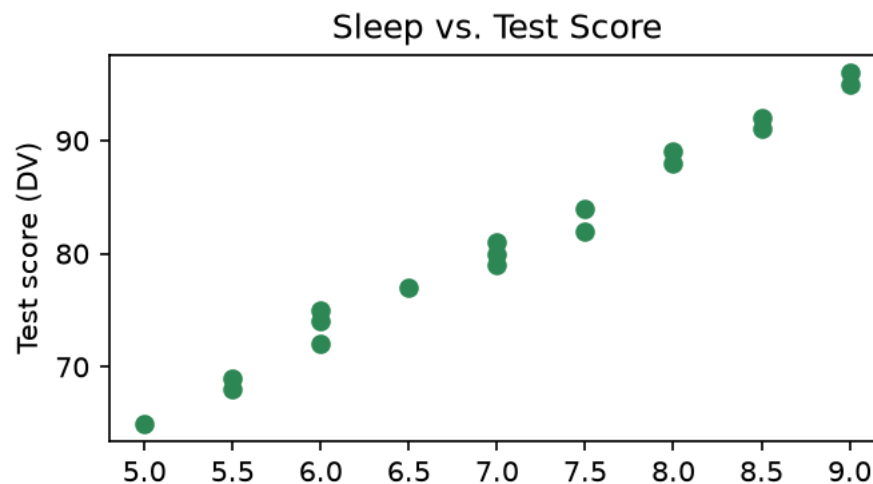


⚠ A **bar chart** counts *categories*; a **histogram** bins *numbers*. Don't mix them up.

Scatter plot – two continuous variables (IV vs DV)

Python R

```
_ = plt.scatter(df["hours_of_sleep"], df["test_score"], color="#2e8b57")  
plt.title("Sleep vs. Test Score")  
plt.xlabel("Hours of sleep (IV)"); plt.ylabel("Test score (DV)")  
plt.show()
```



➤ Each dot is one student. Look for a **trend**: do points drift up, down, or scatter randomly?

! The one rule for every chart

A chart without a title and axis labels is not a chart — it is a picture.

Every chart must be **self-explanatory**: a reader should understand it without you standing next to it.

Every chart needs:

- ✓ A clear **title**
- ✓ A labeled **x-axis** (with units)
- ✓ A labeled **y-axis** (with units)
- ✓ Readable text & legend (if needed)

✂ If you screenshot a chart with no labels for your portfolio, you've saved a picture — not a finding.

AI prompt – generate, run, then polish

1. Ask ChatGPT:

*"Write Python code for a **scatter plot** of **[IV]** vs **[DV]** from my pandas DataFrame *df*."*

2. **Run** it on your real data.

3. **Add** color, a title, and axis labels.

4. **Screenshot** it for your portfolio.

 AI draws fast – **you** turn its picture into a chart.

AI almost always forgets:

- A descriptive **title**
- **Axis labels** with units
- Sensible **color** & point size
- Handling **missing values**

You are the one who makes it self-explanatory.

Team Challenge

Build & label – ~20 min

Using your team data, build:

1. A **histogram** of one continuous variable.
2. A **bar chart** of one categorical variable.
3. A **scatter plot** of your **IV vs DV** .

Every chart must pass the checklist:

- Title ✓
- X-axis label ✓
- Y-axis label ✓

Screenshot each one for your **portfolio**.

 Instructor checks that each chart is **self-explanatory** before you save it.

2:45 – 3:00 · Scientific Storytelling

One sentence under every chart

Add a **one-sentence annotation** below each chart that answers:

- What **pattern** does this show?
- What would a **reader** need to know to interpret it?

 These annotations become the **captions** in your **Results section**.

Describe (Results)

"The bars show higher values in Group A."

Interpret (Discussion)

"Group A reports higher scores, which may reflect..."

Describe now – interpret later

Describing = Results

- States *what the chart shows*
- Just the facts: direction, size, groups
- No "why," no causes
- *"Scores rise as sleep increases."*

Interpreting = Discussion

- Explains *what it might mean*
- Offers reasons, caveats, theory
- Saved for **later** in the paper
- *"...perhaps because rested students focus better."*

 **Results are descriptive.** Save the "why" for the Discussion – mixing them is the most common beginner mistake.

Annotate your three charts – quick round

For **each** of your three charts, write a one-sentence **descriptive** annotation. Then read your scatter-plot annotation aloud to a partner.

Template: *"This [chart type] of [variable(s)] shows [pattern], which means a reader can see [what it tells them]."*

☑ Did you stay **descriptive**? If you wrote the word *"because,"* you slipped into interpretation – move it to the Discussion.

